

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)		identifying 10 from diagram (1) 4 cm (1) shown on chromatogram (ecf) (1) 3 marks for correct dot only	1	1 1		3	3	3
		(ii)		The brown is a combination of yellow (1) and purple / blue / red (1) (so disagree) OR blue (1) and red (1) (so disagree) If a candidate agrees as brown is made of yellow and purple (1) If a candidate agrees as brown is made of {red/blue} and purple (0) Allow: brown has more than one dot (1)			2	2		2
		(iii)		The mobile phase is the solvent (1) The stationary is the paper/plate/covering on plate (1) The dyes separate because they are not equally soluble (1)	3			3		
	(b)	(i)	I	Sodium carbonate and sodium chloride	1			1		1
			II	Sodium carbonate and lithium carbonate	1			1		1
		(ii)		Acidify with nitric acid (1) and add silver nitrate solution (1) White <u>precipitate</u> (1)	3			3		3
				Question 5 total	9	2	2	13	3	10